The Obstruction Spectrum: A Consolidated Theorem–Proof Package

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Attribution. The even/odd dichotomy and the value classification $\{0, d/2\}$ proved in Part I are due to Abramsky, Cercelescu and Constantin (*Commutation Groups and State-Independent Contextuality*, FSCD 2024): their Theorem 16 gives the parity obstruction $2k \equiv 0 \pmod{d}$ and Theorem 21 the normal-form classification. This document is a consolidated, self-contained re-derivation in Weyl-certificate language, together with the tower/valuation material; it does not claim the classification as new.

Abstract

This document collects, in one place and with self-contained proofs, the theorems asserted in the abstract of *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower* ("Paper B") together with the local-valuation theorem of the companion note. Each proof step is explicitly tagged [COMPLETE] (complete self-contained proof), [SKETCH: CITED] (sketch relying on a named external result, cited), or [GAP] (an open sub-lemma). The acceptance target is that every theorem in the Paper B abstract have a proof here, not merely a pointer to a script; where a step is genuinely inherited (notably the general-d attainment of Raussendorf–Okay–Zurel–Feldmann) or genuinely open, we say so. A Status table closes the document. The single most load-bearing external input is the Weyl–Heisenberg product law (Lemma 2), which we derive; everything in Part I is then combinatorial modular arithmetic. The one substantive residual gap is the general even-d *attainment* beyond the residue class $d \equiv 2 \pmod 4$, which we prove in full, and beyond the explicit certificates at $d = 8, 16$: for $d \equiv 0 \pmod 4$ in general we rely on [1].

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1 Conventions and the standing hypotheses

Fix a local dimension $d \geq 2$ and a number of qudits $m \geq 1$. Phase space is $V = \mathbb{Z}^{2m}$ with coordinates written per site $v = (a_1, b_1, \ldots, a_m, b_m)$; we reduce mod d when needed and take representatives in $\{0, \ldots, d-1\}$. Write

$$q(v) = \sum_{i=1}^{m} a_i b_i, \qquad \beta(u,v) = \sum_{i=1}^{m} u_{b,i}\, v_{a,i}, \qquad \langle u, v\rangle = \beta(u,v) - \beta(v,u) = \sum_{i=1}^{m} (u_{a,i} v_{b,i} - u_{b,i} v_{a,i}),$$

so q is a $\mathbb{Z}$-valued quadratic form, β a $\mathbb{Z}$-bilinear form, and $\langle \cdot, \cdot \rangle$ the (alternating, $\mathbb{Z}$-bilinear) symplectic form. Let $\omega = e^{2\pi i/d}$ and $\tau = e^{i\pi/d}$, so $\tau^2 = \omega$, τ has multiplicative order $2d$, and $\tau^d = e^{i\pi} = -1$. On a single qudit $X|j\rangle = |j + 1 \bmod d\rangle$, $Z|j\rangle = \omega^j |j\rangle$; the *symmetric (Weyl) lift* is

$$W(v) = \tau^{-q(v)}\, X^{a_1} Z^{b_1} \otimes \cdots \otimes X^{a_m} Z^{b_m}.$$

Definition 1 (Contexts, families, certificates)**.** A *context* $C = (v_1, \ldots, v_n)$ is a finite word (repetitions allowed) of *pairwise commuting* vectors, $\langle v_i, v_j \rangle \equiv 0 \pmod d$, that is *closed*: $\sum_i v_i \equiv 0 \pmod d$. Its product is a scalar, $\prod_i W(v_i) = \omega^{s(C)} I$ for a well-defined $s(C) \in \mathbb{Z}_d$ (see Lemma 13; for now $s(C)$ names the ω-exponent when it exists). A *family* $\mathcal{F} = \{C_1, \ldots, C_k\}$ has *incidence (multiplicity) matrix* $A \in \mathbb{Z}^{k \times N}$, $A_{r,v}$ = number of times observable v occurs in C_r, over the set of N distinct observables, and *phase vector* $s = (s(C_r))_r \in \mathbb{Z}_d^k$. A *certificate* is $\lambda \in \mathbb{Z}_d^k$ with $\lambda^\mathsf{T} A \equiv 0 \pmod d$; its *value* is $S = \lambda^\mathsf{T} s \bmod d$. A nonzero value certifies that no $\mathbb{Z}_d$-valued noncontextual assignment is consistent with $\mathcal{F}$ (an all-versus-nothing obstruction).

Scope declared once. Throughout Part I the *only* hypotheses on a family are those in Definition 1: each context is closed and internally commuting. We *explicitly permit*: (i) arbitrary multiplicities and repeated observables inside a context (a letter may recur); (ii) nonmaximal contexts (a context need not be a maximal commuting set); (iii) disconnected incidence structures (the observable–context bipartite graph may have several components); (iv) certificates λ with arbitrary $\mathbb{Z}_d$ entries, not only 0/1. Where a result needs more (e.g. "pure fiber pool" in Theorem 18) it is stated locally.

Lemma 2 (Weyl product law)**.** *For all $u, v \in \mathbb{Z}^{2m}$,*

$$W(u)W(v) = \tau^{\,c(u,v)}\, W((u+v) \bmod d), \qquad c(u,v) = -q(u) - q(v) + 2\beta(u,v) + q((u+v) \bmod d),$$

with $c(u,v)$ read modulo $2d$. Consequently $W(u)W(v) = \omega^{\langle u,v\rangle} W(v)W(u)$, so u, v commute iff $\langle u, v\rangle \equiv 0 \pmod d$.

Proof. [COMPLETE] It suffices to treat one qudit and tensor. On one qudit $ZX = \omega XZ$, hence $Z^b X^{a'} = \omega^{a'b} X^{a'} Z^b$ and $X^a Z^b X^{a'} Z^{b'} = \omega^{a'b} X^{a+a'} Z^{b+b'}$. Because $X^d = Z^d = I$, $X^{a+a'} = X^{s_a}$, $Z^{b+b'} = Z^{s_b}$ with $s = (u+v) \bmod d$; and $X^{s_a} Z^{s_b} = \tau^{q(s)} W(s)$. Collecting the τ-powers, using $\omega^{a'b} = \tau^{2a'b} = \tau^{2\beta(u,v)}$ (single qudit $\beta(u,v) = b\,a'$),

$$W(u)W(v) = \tau^{-q(u)-q(v)}\,\tau^{2\beta(u,v)}\,\tau^{q(s)} W(s),$$

which is the claim. Swapping u, v changes the exponent by $2(\beta(u,v) - \beta(v,u)) = 2\langle u, v\rangle$, and $\tau^{2\langle u,v\rangle} = \omega^{\langle u,v\rangle}$. The multi-qudit statement is the tensor product; all three forms $q, \beta, \langle\cdot,\cdot\rangle$ are the site-wise sums, so the identity adds over sites. □

Lemma 3 (Telescoped context phase). *For a context $C = (v_1, \ldots, v_n)$ let $P_i = (v_1 + \cdots + v_{i-1}) \bmod d$ (so $P_1 = 0$). Then $\prod_{i=1}^n W(v_i) = \tau^T W(\sum_i v_i \bmod d)$ with*

$$T = -\sum_i q(v_i) + 2\sum_i \beta(P_i, v_i) \;\equiv\; -\sum_i q(v_i) + 2\sum_{j<i} \beta(v_j, v_i) \pmod{2d}.$$

If C is closed then the product is $\tau^T I$.

Proof. [COMPLETE] Induct with Lemma 2: multiplying the partial product $\tau^{\cdots} W(P_i)$ by $W(v_i)$ adds $c(P_i, v_i) = -q(P_i) - q(v_i) + 2\beta(P_i, v_i) + q(P_{i+1})$, where $P_{i+1} = (P_i + v_i) \bmod d$. Summing $i = 1, \ldots, n$, the pair $-q(P_i) + q(P_{i+1})$ telescopes to $q(P_{n+1}) - q(P_1) = q(0) - q(0) = 0$. This leaves $T = -\sum_i q(v_i) + 2\sum_i \beta(P_i, v_i)$. Finally $\beta(P_i, v_i) \equiv \beta(\sum_{j<i} v_j, v_i) = \sum_{j<i} \beta(v_j, v_i) \pmod d$ by $\mathbb{Z}$-bilinearity (replacing P_i by the unreduced prefix changes β by a multiple of d), so $2\sum_i \beta(P_i, v_i) \equiv 2\sum_{j<i} \beta(v_j, v_i) \pmod{2d}$. □

Part I. The Obstruction Spectrum

2 The Obstruction Spectrum Theorem

Theorem 4 (Obstruction Spectrum). *For every d, every m, every family $\mathcal{F}$ satisfying Definition 1, and every certificate λ, the value obeys*

$$2S \equiv 0 \pmod d.$$

Hence $S \in \{0, d/2\}$ for even d, and $S = 0$ for odd d. Thus the set $H(d,m)$ of achievable state-independent values satisfies $H \subseteq \{0, d/2\}$ (even d) and $H = \{0\}$ (odd d). The bound holds verbatim in the presence of multiplicities, repeated observables, nonmaximal contexts, and disconnected incidence.

Proof. [COMPLETE] Fix a closed commuting context C. Reduce the exact identity of Lemma 3 modulo d:

$$2s(C) \equiv -\sum_i q(v_i) + 2\sum_{j<i} \beta(v_j, v_i) \pmod d. \tag{1}$$

(Here $\tau^T = \omega^{s(C)}$ means $2s(C) \equiv T \pmod{2d}$; we use it mod d.) Now set $\sigma = \sum_i v_i$. Bilinearity gives

$$\beta(\sigma, \sigma) = \sum_i \beta(v_i, v_i) + \sum_{i<j} (\beta(v_i, v_j) + \beta(v_j, v_i)) = \sum_i q(v_i) + \sum_{i<j} (\beta(v_i, v_j) + \beta(v_j, v_i)),$$

using $\beta(v,v)=q(v)$. On commuting pairs $\beta(v_i,v_j)\equiv\beta(v_j,v_i) \pmod d$ (because their difference is $\langle v_i,v_j\rangle\equiv 0$), so $\beta(\sigma,\sigma)\equiv\sum_i q(v_i)+2\sum_{i<j}\beta(v_j,v_i) \pmod d$. Closedness gives $\sigma\equiv 0 \pmod d$, i.e. $\sigma=d\,t$ for an integer vector t, whence $\beta(\sigma,\sigma)=d^2\beta(t,t)\equiv 0 \pmod d$. Therefore

$$2\sum_{i<j}\beta(v_j,v_i)\equiv-\sum_i q(v_i) \pmod d. \tag{2}$$

Substituting (2) into (1),

$$2s(C)\equiv-2\sum_{v\in C}q(v) \pmod d. \tag{3}$$

Equation (3) says $2s\equiv-2Aq \pmod d$ as vectors over contexts, where $q=(q(v))_v$. For any certificate ($\lambda^{\mathsf T}A\equiv 0$),

$$2S=\lambda^{\mathsf T}(2s)\equiv-2\,(\lambda^{\mathsf T}A)\,q\equiv 0 \pmod d.$$

For even d, $2S\equiv 0$ forces $S\in\{0,d/2\}$; for odd d, 2 is invertible so $S=0$. None of the steps used maximality, distinctness of observables, connectivity, or a bound on multiplicities; C was an arbitrary closed commuting word and λ an arbitrary left-kernel vector. □

Corollary 5 (Odd d is trivial, sharply). *For odd d, dividing (3) by the invertible 2 gives $s(C)\equiv-\sum_{v\in C}q(v) \pmod d$ outright, so $s\equiv-Aq$ is an exact coboundary and every certificate value is 0.* [COMPLETE]

Corollary 6 (The obstruction is one bit). *For even d, (3) gives $s(C)+\sum_{v\in C}q(v)\equiv 0 \pmod{d/2}$, so*

$$b(C):=\frac{s(C)+\sum_{v\in C}q(v)}{d/2}\bmod 2\ \in\mathbb{Z}_2$$

is well defined, and $s(C)\equiv-\sum_{v\in C}q(v)+\frac{d}{2}b(C) \pmod d$. For any certificate, $S\equiv\frac{d}{2}(\bar\lambda\cdot b) \pmod d$ with $\bar\lambda=\lambda\bmod 2$; only $\lambda\bmod 2$ carries value, and an attaining certificate must have odd $\bar\lambda\cdot b$. [COMPLETE]

Proof. [COMPLETE] $b(C)$ is well defined by (3). For a certificate, $S=\lambda^{\mathsf T}s\equiv-\lambda^{\mathsf T}Aq+\frac{d}{2}\lambda^{\mathsf T}b\equiv\frac{d}{2}(\lambda\cdot b) \pmod d$, and $\frac{d}{2}(\lambda\cdot b)\bmod d$ depends only on $\lambda\cdot b\bmod 2=\bar\lambda\cdot b$. □

Remark 7 (No obstruction of order >2). Corollary 6 shows the value always lies in the order-2 subgroup $\{0,d/2\}\subset\mathbb{Z}_d$, *whatever* the multiplicity structure. If each observable lies in r contexts, the value group is still $\subseteq\{0,d/2\}$, independent of r: there are no genuine order-d "qudit parity proofs". [COMPLETE]

3 The commutator–carry form

Theorem 8 (Carry form). *Exactly, $2s(C)\equiv-\sum_i q(v_i)+2\sum_{j<i}\beta(v_j,v_i) \pmod{2d}$ (Lemma 3), and for even d the obstruction bit of Corollary 6 equals the total commutator carry:*

$$b(C)\equiv\sum_{\{u,v\}\subseteq C}\frac{\langle u,v\rangle}{d} \pmod 2,$$

the sum over unordered pairs of positions in the word C. The entire even-d obstruction theory is thus combinatorial: no phases, no matrices, only symplectic carries $\langle u,v\rangle/d\in\mathbb{Z}$.

Proof. [COMPLETE] We derive the carry expression from the cocycle formula rather than assert it. Start from the polarization computed inside Theorem 4, but keep the symmetric/antisymmetric split exactly. Using $\beta(v_i, v_j) = \beta(v_j, v_i) + \langle v_i, v_j\rangle$,

$$\beta(\sigma,\sigma) = \sum_i q(v_i) + \sum_{i<j}\big(2\beta(v_j,v_i) + \langle v_i, v_j\rangle\big) = \Big[\sum_i q(v_i) + 2\sum_{i<j}\beta(v_j,v_i)\Big] + \sum_{i<j}\langle v_i, v_j\rangle.$$

Hence, with $\sigma = d\,t$ so $\beta(\sigma,\sigma) = d^2\beta(t,t)$,

$$\sum_i q(v_i) + 2\sum_{i<j}\beta(v_j,v_i) = d^2\beta(t,t) - \sum_{i<j}\langle v_i, v_j\rangle. \tag{4}$$

Now compute $b(C)$ from its definition. By Corollary 6, $\frac{d}{2}\,b(C) \equiv s(C) + \sum q(v_i) \pmod d$, so $d\,b(C) \equiv 2s(C) + 2\sum q(v_i) \pmod{2d}$, and by (1) lifted to the exact mod-$2d$ identity of Lemma 3,

$$d\,b(C) \equiv \Big(-\sum q(v_i) + 2\sum_{j<i}\beta(v_j,v_i)\Big) + 2\sum q(v_i) = \sum q(v_i) + 2\sum_{i<j}\beta(v_j,v_i) \pmod{2d}.$$

Insert (4): $d\,b(C) \equiv d^2\beta(t,t) - \sum_{i<j}\langle v_i, v_j\rangle \pmod{2d}$. Each commuting pair has $\langle v_i, v_j\rangle \equiv 0 \pmod d$, so $\langle v_i, v_j\rangle/d \in \mathbb{Z}$ and we may divide by d:

$$b(C) \equiv d\,\beta(t,t) - \sum_{i<j}\frac{\langle v_i, v_j\rangle}{d} \pmod 2.$$

For even d the term $d\,\beta(t,t) \equiv 0 \pmod 2$, and $-1 \equiv 1 \pmod 2$, giving $b(C) \equiv \sum_{i<j}\langle v_i, v_j\rangle/d \pmod 2$. Diagonal "pairs" from a repeated letter contribute $\langle v, v\rangle/d = 0$, so the formula is insensitive to repetitions, as claimed. □

Corollary 9 (Closed-form contextuality criterion; cycle = self-pairing)**.** *Let d be even and let A (over $\mathbb{F}_2$) be the incidence matrix, $b = (b(C_r))_r$ the carry vector. Call a $\mathbb{Z}_2$ left-kernel vector λ ($\lambda^\mathsf{T} A \equiv 0 \bmod 2$) a* cycle*. Then*

$$\lambda\cdot b \;\equiv\; Q(\lambda) := \sum_{a<b \textit{ in the observable multiset of } \lambda} \frac{\langle v_a, v_b\rangle}{d} \pmod 2,$$

and the family carries a $\mathbb{Z}_2$ all-versus-nothing obstruction iff some cycle has $Q(\lambda)$ odd.

Proof. [COMPLETE] Write $Q(\lambda) = \text{intra} + \text{inter}$ over pairs inside a single selected context and pairs across two selected contexts. By definition $\text{intra} = \sum_{r:\lambda_r=1}\sum_{i<j\in C_r}\langle v_i, v_j\rangle/d = \lambda\cdot b$ (Theorem 8). For the inter term, bilinearity gives $\sum_{u\in C_a, v\in C_b}\langle u, v\rangle = \langle S_a, S_b\rangle$ with $S_r = \sum_{v\in C_r} v = d\,t_r$ (closedness), so $\langle S_a, S_b\rangle = d^2\langle t_a, t_b\rangle$ and $\text{inter} = \sum_{a<b} d\langle t_a, t_b\rangle \equiv 0 \pmod 2$ for even d. Hence $Q(\lambda) = \lambda\cdot b$. The obstruction is $b \notin \text{rowspace}_{\mathbb{F}_2}(A)$, i.e. some left-kernel λ pairs oddly with b. □

Proposition 10 (The exact $\mathbb{Z}_d$ criterion; Q is its mod-2 shadow)**.** *Let M be the $\mathbb{Z}_d$ incidence matrix and $\gamma \in \mathbb{Z}_d^k$ the context-scalar vector. By the Fredholm alternative over $\mathbb{Z}_d$ (Lemma 11), $\mathcal{F}$ is state-independent contextual iff some $\mathbb{Z}_d$-cycle λ ($\lambda^\mathsf{T} M \equiv 0 \bmod d$) has $\lambda\cdot\gamma \not\equiv 0 \pmod d$. The odd-$Q$ test of Corollary 9 is the mod-2 reduction: a $\mathbb{Z}_d$-cycle reduces to a $\mathbb{Z}_2$-cycle (so odd-Q is* sound*), but a $\mathbb{Z}_2$-cycle need not lift, so odd-Q can be* incomplete *on engineered minimal supports.* [COMPLETE] *for soundness; the empirical faithfulness on* Weyl *families (that the $\mathbb{Z}_2$ shadow and the $\mathbb{Z}_d$ obstruction agree except on engineered supports) is a verified phenomenon, not proved here* [GAP].

4 Detection equivalence

We first isolate the pure linear algebra, then feed in the Spectrum Theorem.

Lemma 11 (Fredholm alternative / double annihilator over $\mathbb{Z}_d$)**.** *Let $A \in \mathbb{Z}^{k\times N}$ and $s \in \mathbb{Z}^k$. Exactly one of the following holds:*

(a) there is $\gamma \in \mathbb{Z}_d^N$ with $A\gamma \equiv s \pmod d$ (a noncontextual assignment)*;*

(b) there is $\lambda \in \mathbb{Z}_d^k$ with $\lambda^\mathsf{T} A \equiv 0 \pmod d$ and $\lambda^\mathsf{T} s \not\equiv 0 \pmod d$ (a certificate).

Moreover in case (b) an explicit λ is produced constructively from the Smith Normal Form, valid for composite *d.*

Proof. [COMPLETE] *Mutual exclusion.* If both held, $\lambda^\mathsf{T} s \equiv \lambda^\mathsf{T} A\gamma \equiv 0$, contradiction. *Exhaustiveness with construction.* Compute the Smith Normal Form over $\mathbb{Z}$: $UAV = D = \mathrm{diag}(g_1, \dots, g_r, 0, \dots, 0)$ with $U \in \mathrm{GL}_k(\mathbb{Z})$, $V \in \mathrm{GL}_N(\mathbb{Z})$, $g_1 \mid \cdots \mid g_r$. Put $s' = Us$. Since U, V are unimodular they are invertible mod d, so $A\gamma \equiv s \pmod d$ is solvable iff $D\gamma' \equiv s' \pmod d$ is (with $\gamma' = V^{-1}\gamma$), which decouples into $g_i\gamma_i' \equiv s_i' \pmod d$ $(i \le r)$ and $0 \equiv s_i' \pmod d$ $(i > r)$. The scalar congruence $g_i\gamma_i' \equiv s_i' \pmod d$ is solvable iff $\gcd(g_i, d) \mid s_i'$; the $i > r$ rows are solvable iff $d \mid s_i'$ (the case $\gcd(0, d) = d$).

If (a) fails, some index i violates its condition. Set $\mu^\mathsf{T} = \dfrac{d}{\gcd(g_i, d)}\, e_i^\mathsf{T}$ (with $g_i = 0$ read as $\gcd = d$, i.e. $\mu^\mathsf{T} = e_i^\mathsf{T}$), and $\lambda^\mathsf{T} = \mu^\mathsf{T} U$. Then $\lambda^\mathsf{T} A = \mu^\mathsf{T} UA = \mu^\mathsf{T} DV^{-1}$, and $\mu^\mathsf{T} D = \frac{d}{\gcd(g_i,d)} g_i\, e_i^\mathsf{T} = d \cdot \frac{g_i}{\gcd(g_i,d)}\, e_i^\mathsf{T} \equiv 0 \pmod d$, so $\lambda^\mathsf{T} A \equiv 0$. And $\lambda^\mathsf{T} s = \mu^\mathsf{T} s' = \frac{d}{\gcd(g_i,d)} s_i'$, which is $\not\equiv 0 \pmod d$ precisely because $\gcd(g_i, d) \nmid s_i'$. This is a certificate, giving (b). □

Remark 12 (Why $d/\gcd(g_i, d)$ and not d/g_i). The invariant factor g_i need *not* divide d; writing $u = (d/g_i)U_i$ would be ill-formed for composite d. The correct annihilator multiplier is $d/\gcd(g_i, d)$, which coincides with d/g_i only when $g_i \mid d$. [COMPLETE]

Lemma 13 (Integrality/gauge)**.** *For even d and a closed commuting context C, the product $\prod_{v\in C} W(v)$ is an ω-power $\omega^{s(C)}I$ with $s(C) \in \mathbb{Z}_d$* canonically *(no τ-half-steps). Kernel pairings $\lambda^\mathsf{T} s$ are independent of any admissible regauging of the $W(v)$.*

Proof. [COMPLETE] By Lemma 3 the product is $\tau^T I$ with $T = -\sum q(v_i) + 2\sum_{j<i} \beta(v_j, v_i)$. By the parity lemma (Lemma 26 of Part II) at even d every commuting pair has $c(v_j, v_i)$ even; telescoping, T is even, so $\tau^T = \omega^{T/2}$ and $s(C) = T/2 \bmod d \in \mathbb{Z}_d$. A regauging $W(v) \mapsto \zeta_v W(v)$ with $\zeta_v \in U(1)$ shifts $s(C) \mapsto s(C) + \sum_{v\in C} \theta_v$ $(\zeta_v = \omega^{\theta_v})$, i.e. $s \mapsto s + A\theta$; hence $\lambda^\mathsf{T} s \mapsto \lambda^\mathsf{T} s + (\lambda^\mathsf{T} A)\theta = \lambda^\mathsf{T} s$ for a certificate. □

Theorem 14 (Detection equivalence)**.** *Let d be even and $\mathcal{F}$ a closed commuting Weyl family with incidence matrix A (over $\mathbb{Z}_d$) and canonical phase vector $s \in \mathbb{Z}_d^k$ (Lemma 13). The following are equivalent:*

(a) no noncontextual $\mathbb{Z}_d$-assignment exists (the system $A\gamma \equiv s \bmod d$ is infeasible);

(b) some certificate λ ($\lambda^\mathsf{T} A \equiv 0$) has $\lambda^\mathsf{T} s \equiv d/2 \pmod d$;

(c) some certificate has odd commutator carry, $\bar\lambda \cdot b \equiv 1 \pmod 2$.

Proof. [COMPLETE] (a)⇔"$\exists$ certificate with $\lambda^{\mathsf{T}} s \not\equiv 0$" is exactly Lemma 11 with the identifications $M = A$, phase $= s$. Given such a certificate, the Obstruction Spectrum Theorem 4 (which applies to *any* left-kernel vector, by the scope declared in §1 and Remark 7) forces $2(\lambda^{\mathsf{T}} s) \equiv 0$, hence the nonzero value is exactly $d/2$: this is (b). Trivially (b)⇒(a) by mutual exclusion in Lemma 11. Finally (b)⇔(c) is Corollary 6: $\lambda^{\mathsf{T}} s \equiv \frac{d}{2}(\bar{\lambda} \cdot b)$, which is $d/2$ iff $\bar{\lambda} \cdot b$ is odd. The canonical integral gauge needed to even state (a) is Lemma 13; kernel pairings are gauge invariant, so the equivalence is gauge free. □

Remark 15 (What is new here vs. assembly). The three-way equivalence is five lines of ring theory (Lemma 11) bolted onto two proven theorems of Part I. The composite-d subtlety — that Fredholm over $\mathbb{Z}_d$ needs SNF and the $d/\gcd(g_i, d)$ multiplier — is genuine and fully discharged above. [COMPLETE]

5 The exact value-bit formula

We now promote the computational value-bit identity to a theorem. The setting is a *pure fiber pool*: a base family F at level d and, at level $2d$, observables $\tilde{v} = u + d\,x$ with $u = \tilde{v} \bmod d \in \{0, \ldots, d-1\}^{2m}$ the base residue and $x = \tilde{v} \text{ div } d \in \{0,1\}^{2m}$ the lift bits; a mod-$2d$ kernel element $\tilde{\lambda}$ with odd support supp.

Lemma 16 (Sum-of-floors carry identity)**.** *For integers $M_1, \ldots, M_n$ with $\Sigma = \sum_a M_a$ and $\kappa = \#\{a : M_a \text{ odd}\}$,*

$$\sum_a \left\lfloor \frac{M_a}{2} \right\rfloor \equiv \left\lfloor \frac{\Sigma}{2} \right\rfloor + \left\lfloor \frac{\kappa}{2} \right\rfloor \pmod 2.$$

Proof. [COMPLETE] $\lfloor M/2 \rfloor = (M - (M \bmod 2))/2$ gives $\sum_a \lfloor M_a/2 \rfloor = (\Sigma - \kappa)/2$. Since $\Sigma \equiv \kappa \pmod 2$, $\lfloor \Sigma/2 \rfloor = (\Sigma - (\kappa \bmod 2))/2$, and the difference is $\big((\kappa \bmod 2) - \kappa\big)/2 = -\lfloor \kappa/2 \rfloor \equiv \lfloor \kappa/2 \rfloor \pmod 2$. □

Lemma 17 (Exact per-pair lift carry)**.** *With $q_u = \lfloor \langle u_a, u_b \rangle / d \rfloor$ replaced by $\langle u_a, u_b \rangle \bmod d \in [0, d)$ in the numerator, write $M_{ab} = \langle u_a, x_b \rangle + \langle x_a, u_b \rangle$, $s_{xx} = \langle x_a, x_b \rangle$. Then for a level-$2d$ commuting pair,*

$$\left\lfloor \frac{\langle \tilde{v}_a, \tilde{v}_b \rangle}{2d} \right\rfloor \equiv \left\lfloor \frac{(\langle u_a, u_b \rangle / d) + M_{ab} + d\, s_{xx}}{2} \right\rfloor \pmod 2,$$

a pure floor identity because $0 \le \langle u_a, u_b \rangle \bmod d < d$ prevents the base remainder from carrying.

Proof. [SKETCH: CITED] Expand $\langle \tilde{v}_a, \tilde{v}_b \rangle = \langle u_a, u_b \rangle + dM_{ab} + d^2 s_{xx}$; divide by $2d$ and use that the pair commutes at level $2d$ (numerator $\equiv 0 \bmod 2d$) so the fraction is an integer, and that $\langle u_a, u_b \rangle = d \lfloor \langle u_a, u_b \rangle / d \rfloor + (\langle u_a, u_b \rangle \bmod d)$ with the remainder in $[0, d)$. The remainder cannot contribute a $2d$-carry; the residual is the displayed floor. The boundary bookkeeping is verified exactly at $d = 4, 6, 8$ in `Wformula.py`. *Cited:* Paper B, App. A. □

Theorem 18 (Value-bit formula)**.** *For a mod-$2d$ kernel element $\tilde{\lambda}$ of a pure fiber pool with odd support* supp*, the value bit P (defined by $S = d\,P$) is*

$$P \equiv \frac{1}{2} \Big[\sum_{\text{supp}} K + \sum_{\text{supp}} \Lambda(x) + \frac{d}{2} \sum_{\text{supp}} Q(x) \Big] \pmod 2,$$

where, per selected context, $K = \sum_{i<i'} \langle u_i, u_{i'} \rangle / d$, $\Lambda(x) = \langle \sum_i x_i, \sigma \rangle - \sum_i \langle x_i, u_i \rangle - 2 \sum_i \langle x_i, P_i \rangle$, $Q(x) = \sum_{i<i'} \langle x_i, x_{i'} \rangle$, $\sigma = \sum_i u_i$, P_i the base prefix sums. The bracket is always even, and the $(d/2)$-terms survive mod 2 only for $d \equiv 2 \pmod 4$.

Proof. [SKETCH: CITED] By Corollary 6 at level $2d$, $P = \bar{\bar{\lambda}} \cdot b_{2d} = \sum_{C\in\text{supp}} b_{2d}(C) \bmod 2$, and by Theorem 8 $b_{2d}(C) = \sum_{i<i'} \langle \tilde{v}_i, \tilde{v}_{i'}\rangle/(2d)$. Lemma 17 rewrites each summand as $\lfloor((\langle u_i, u_{i'}\rangle/d) + M_{ii'} + d\, s_{ii'})/2\rfloor$; Lemma 16 then converts the sum of per-pair floors into (i) $\lfloor(\sum_{i<i'}(\ldots))/2\rfloor$ and (ii) a parity-count carry $\lfloor\kappa/2\rfloor$. Grouping the base term into K, the cross term $\sum_{i<i'} M_{ii'}$ into Λ (the identity $\sum_{i<i'} M_{ii'} = \langle\sum x_i, \sigma\rangle - \sum\langle x_i, u_i\rangle - 2\sum\langle x_i, P_i\rangle$ is bilinearity plus the prefix expansion), and the quadratic term $\sum s_{ii'}$ into $Q(x)$ with the explicit $d/2$ weight, yields the displayed bracket. Bracket evenness is equivalent to $P \in \mathbb{Z}$, which holds because $S \in \{0, d\}$ at level $2d$ (Theorem 4). The exact matching of the residual carry $\lfloor\kappa/2\rfloor$ against the stated coefficients is the content verified non-vacuously (0/550) in `Wformula.py`. *Cited:* Paper B, Thm. W. □

Status of the step. Structure (reduction to $\sum b_{2d}$ and the three natural terms) is [COMPLETE]; the exact integer coefficients and bracket-evenness rest on Lemma 17 and are computationally certified, hence [SKETCH: CITED] overall. This is the one Part-I theorem not fully closed by hand here.

6 Attainment and the classification

Theorem 19 (Attainment / classification)**.** *For every even d and every $m \geq 2$, the value $d/2$ is attained by a closed commuting Weyl family; combined with Theorem 4, $H(d, m) = \{0, d/2\}$ for even d and $H(d, m) = \{0\}$ for odd d.*

We prove attainment in three pieces of decreasing self-containment, and label each honestly.

(A) A complete parametric construction for $d \equiv 2 \pmod 4$

Proposition 20 (Signed scaled Peres–Mermin)**.** *Let $d \equiv 2 \pmod 4$ and $\delta = d/2$ (odd). Let $\{v_1, \ldots, v_9\} \subset \mathbb{Z}_2^4$ be the nine Peres–Mermin vectors on $m = 2$ qudits arranged in a 3×3 grid, with the six line-contexts (three rows, three columns). Scale every vector to $\delta v \in \mathbb{Z}_d^4$. Then the six scaled lines are closed commuting contexts, and the signed vector $\lambda = (+1, +1, +1, -1, -1, -1)$ (rows $+1$, columns -1) is a genuine $\mathbb{Z}_d$-certificate with value $S \equiv d/2$.*

Proof. [COMPLETE] *Grid.* Take, in per-site coordinates (a_1, b_1, a_2, b_2),

$$\begin{array}{ccc} X\otimes I = (1,0,0,0) & I\otimes X = (0,0,1,0) & X\otimes X = (1,0,1,0) \\ I\otimes Z = (0,0,0,1) & Z\otimes I = (0,1,0,0) & Z\otimes Z = (0,1,0,1) \\ X\otimes Z = (1,0,0,1) & Z\otimes X = (0,1,1,0) & Y\otimes Y = (1,1,1,1), \end{array}$$

rows and columns being the six contexts. Each of the six lines sums to an even vector ($(2,0,2,0)$, $(0,2,0,2)$, $(2,2,2,2)$, $(2,0,0,2)$, $(0,2,2,0)$, $(2,2,2,2)$), and within each line every pair has *even* integer symplectic product (the operators pairwise commute over $\mathbb{Z}_2$). Each of the nine observables lies in exactly one row and one column.

Closure and commutation after scaling. $\sum_{v\in\text{line}} \delta v = \delta \cdot (\text{even vector}) = d \cdot (\text{integer vector}) \equiv 0 \pmod d$; and for u, w in a common line $\langle \delta u, \delta w\rangle = \delta^2 \langle u, w\rangle = \frac{d^2}{4}\langle u, w\rangle$; since $\langle u, w\rangle$ is even, write $\langle u, w\rangle = 2w_{uw}$, so $\langle \delta u, \delta w\rangle = \frac{d^2}{2} w_{uw} = d \cdot (\delta w_{uw}) \equiv 0 \pmod d$. So the scaled lines are closed commuting contexts.

Certificate. Each observable is one row-edge and one column-edge of $K_{3,3}$, so $(\lambda^{\mathsf{T}} A)_v = \lambda_{\text{row}(v)} + \lambda_{\text{col}(v)} = (+1) + (-1) = 0$ exactly in $\mathbb{Z}_d$; λ is a genuine $\mathbb{Z}_d$-cycle.

Value. By Corollary 6, $s(C) \equiv -\sum_{v\in C} q(v) + \frac{d}{2} b(C)$, so

$$S = \lambda^{\mathsf{T}} s = \sum_{\text{rows}} s - \sum_{\text{cols}} s \equiv -\Big(\sum_{\text{rows}}\sum_{v\in C} q(v) - \sum_{\text{cols}}\sum_{v\in C} q(v)\Big) + \frac{d}{2}\Big(\sum_{\text{rows}} b - \sum_{\text{cols}} b\Big) \pmod d.$$

Each observable appears once among rows and once among columns, so the two $\sum q$ totals are both $\sum_{\text{obs}} q(v)$ and cancel. For the carry, compute $b(\delta C)$: with $\langle \delta u, \delta w\rangle = \frac{d^2}{2} w_{uw}$ and $\langle \delta u, \delta w\rangle / d = \delta w_{uw}$,

$$b(\delta C) \equiv \sum_{u<w\in C} \frac{\langle \delta u, \delta w\rangle}{d} = \delta \sum_{u<w\in C} w_{uw} \equiv \sum_{u<w\in C} w_{uw} \pmod 2 \qquad (\delta \text{ odd}),$$

and $w_{uw} = \langle u, w\rangle/2$ is exactly the $d = 2$ carry summand, so $b(\delta C) \equiv b_{\text{PM}}(C) \pmod 2$ context-by-context. Because the qubit Peres–Mermin square is contextual with value 1, the all-lines total carry is odd: $\sum_{\text{all } C} b_{\text{PM}}(C) \equiv 1$. Hence $\sum_{\text{rows}} b - \sum_{\text{cols}} b \equiv \sum_{\text{all}} b \equiv 1 \pmod 2$, and $S \equiv \frac{d}{2} \cdot 1 = d/2 \pmod d$. □

This is a fully self-contained attainment at *every* $d \equiv 2 \pmod 4$ ($d = 2, 6, 10, 14, \dots$), with an explicit genuine $\mathbb{Z}_d$-certificate.

(B) General even d: inherited from ROZF

Proposition 21 (General even-d attainment)**.** *For every even d (in particular $d \equiv 0 \pmod 4$, where construction (A) degenerates) there is a closed commuting Weyl family attaining $d/2$.*

Proof. [SKETCH: CITED] The elementary scaling of (A) fails for $4 \mid d$: then $\delta = d/2$ is even, every scaled carry $b(\delta C) \equiv \delta \sum w_{uw} \equiv 0 \pmod 2$ vanishes, and the family is noncontextual — a genuinely different construction is required. Raussendorf–Okay–Zurel–Feldmann [1] construct, for every even d, an even-d cohomological ("face") obstruction realizing the value $d/2$; their face construction is reported to attain $\{0, d/2\}$ at $d = 4, 6, 8, 10, 12, 16, 20$. We *inherit* the generality of attainment from [1] and only translate their data into the present certificate language (λ, A, s), where the value $d/2$ then follows from Corollary 6. *Cited:* [1], Quantum **7**, 979 (2023). □

Status of the step. Generality over all even d is [SKETCH: CITED], credited to [1]. What is *original* here: (i) the Weyl-certificate translation of the obstruction into (λ, A, s) with the value read off Corollary 6; (ii) the closed parametric family of (A) for $d \equiv 2 \pmod 4$; and (iii) the explicit certificates of (C). What is *inherited*: the existence of an attaining family at $d \equiv 0 \pmod 4$ in general.

(C) Explicit original certificates at $d = 8, 16$

Proposition 22 (Matrix-verified certificates)**.** *There is an explicit closed commuting Weyl family at $d = 8$ (89 contexts) with a certificate of value $4 = d/2$, and one at $d = 16$ (363 contexts) with value $8 = d/2$. Both lie in the class $d \equiv 0 \pmod 4$ that construction (A) cannot reach.*

Proof. [SKETCH: CITED] The certificates are exhibited and checked on raw 64×64 and 256×256 complex matrices: every listed context is verified pairwise commuting with scalar product, $\lambda^{\mathsf{T}} A \equiv 0 \pmod d$ holds, the value equals $d/2$, and the noncontextual system is infeasible (`verify_cert8.py`, `verify_cert16.py`). These certificates are new to this program and are independent of the tower results; the $d = 16$ instance shows there is no depth ceiling. The by-hand reduction of these finite objects to Corollary 6 is routine but the data are verified by machine, hence [SKETCH: CITED]. *Cited:* Paper B, §6. [COMPLETE] for the *logic* that a value-$d/2$ certificate implies attainment (Theorem 4, Corollary 6); [SKETCH: CITED] for the certificates' numerical correctness. □

7 Depth Rigidity

Definition 23 (Lift; "top 2-adic layer")**.** A *lift* of a level-d family to level $2d$ assigns $x_v \in \{0,1\}^{2m}$ per observable, setting $\tilde{v} = v + d\,x_v$, subject to keeping every context commuting and closed at level $2d$. The mod-2 carry class $b \in \mathbb{F}_2^k/\operatorname{rowspace}_{\mathbb{F}_2} A$ is the *top 2-adic layer* of the family: it is the obstruction visible at the current level and *not* inherited from any finer datum one level up. "No single lift carries the obstruction" means: the whole b-class is a coboundary of a per-observable function whenever any global lift exists, and conversely a genuine b-class forbids every global lift.

Theorem 24 (Depth Rigidity)**.** *Fix even d and a closed commuting level-d family. Reducing the commutation constraints at level $2d$ gives, for each in-context pair (i,j), the $\mathbb{F}_2$-linear equation*

$$\langle v_i, x_j\rangle + \langle x_i, v_j\rangle \equiv k_{ij} := \frac{\langle v_i, v_j\rangle}{d} \pmod 2, \tag{5}$$

and the closure equation $r_C + \sum_{v\in C} x_v \equiv 0 \pmod 2$ with $r_C = \frac{1}{d}\sum_{v\in C} v$.

- *(a) If the commutation block (5) is solvable, then $b(C) \equiv \sum_{v\in C}\langle v, x_v\rangle \pmod 2$ is a coboundary $b = Ah$ ($h_v = \langle v, x_v\rangle \bmod 2$); hence every certificate of the lifted family has value 0.*
- *(b) If the level-d family attains ($\exists\lambda$ with value $d/2$), the $\bar{\lambda}$-weighted commutation equations have identically zero left-hand side and right-hand side $\sum_r \bar{\lambda}_r b(C_r) \equiv 1$; the block is infeasible, so* attaining families never lift*.*

Proof. [COMPLETE] *Derivation of* (5). $\langle \tilde{v}_i, \tilde{v}_j\rangle = \langle v_i, v_j\rangle + d(\langle v_i, x_j\rangle + \langle x_i, v_j\rangle) + d^2\langle x_i, x_j\rangle$. Commuting at level d gives $\langle v_i, v_j\rangle = d\,k_{ij}$; dividing $\langle \tilde{v}_i, \tilde{v}_j\rangle \equiv 0 \pmod{2d}$ by d and using $d^2\langle x_i, x_j\rangle \equiv 0 \pmod 2$ (even d) gives $k_{ij} + \langle v_i, x_j\rangle + \langle x_i, v_j\rangle \equiv 0 \pmod 2$, i.e. (5). Closure is identical with $\sum \tilde{v} = d(r_C + \sum x_v) \equiv 0 \pmod{2d}$.

A per-context reduction. Work mod 2, where $\langle a, b\rangle \equiv \langle b, a\rangle$. For any context,

$$\sum_{i<j}(\langle v_i, x_j\rangle + \langle x_i, v_j\rangle) \equiv \sum_{i\neq j}\langle v_i, x_j\rangle = \langle \textstyle\sum_i v_i, \sum_j x_j\rangle - \sum_i\langle v_i, x_i\rangle \pmod 2,$$

and $\sum_i v_i = d\,r_C$ makes $\langle \sum v_i, \sum x_j\rangle = d\langle r_C, \sum x_j\rangle \equiv 0 \pmod 2$. Hence

$$\sum_{i<j\in C}(\langle v_i, x_j\rangle + \langle x_i, v_j\rangle) \equiv \sum_{v\in C}\langle v, x_v\rangle \pmod 2. \tag{6}$$

(a). If (5) holds for a solution x, sum it over $i<j\in C$: the left side is (6) $= \sum_{v\in C}\langle v, x_v\rangle = (Ah)_C$ with $h_v = \langle v, x_v\rangle \bmod 2$, while the right side is $\sum_{i<j\in C} k_{ij} = \sum_{i<j}\langle v_i, v_j\rangle/d = b(C)$ (Theorem 8). Thus $b = Ah$, a coboundary; any certificate $\bar{\lambda}$ ($\bar{\lambda}^\mathsf{T} A \equiv 0$) pairs to $\bar{\lambda}\cdot b = \bar{\lambda}^\mathsf{T} Ah = 0$, so all lifted values vanish.

(b). Take an attaining λ; then $\bar{\lambda}\cdot b \equiv 1$ (Corollary 6). Weight (5) by $\bar{\lambda}_C$ and sum over all contexts and in-context pairs. The right side is $\sum_C \bar{\lambda}_C \sum_{i<j\in C} k_{ij} = \sum_C \bar{\lambda}_C b(C) = \bar{\lambda}\cdot b \equiv 1$. The left side, by (6), is $\sum_C \bar{\lambda}_C \sum_{v\in C}\langle v, x_v\rangle = \sum_v (\bar{\lambda}^\mathsf{T} A)_v\langle v, x_v\rangle \equiv 0$, because $\lambda^\mathsf{T} A \equiv 0 \pmod d$ reduces to $\bar{\lambda}^\mathsf{T} A \equiv 0 \pmod 2$. So any x would give $0 \equiv 1$: the commutation block is infeasible and no lift exists. $\square$

Remark 25 (Reading of the rigidity). The obstruction bit is precisely the $\mathbb{F}_2$-inconsistency of the global lift system, pinned by the *certificate* $\bar{\lambda}$ and not by any observable's individual lift x_v (each x_v enters only through the coboundary $h_v = \langle v, x_v\rangle$, which pairs to 0 with every cycle). Whenever a global lift exists the obstruction trivializes; when it does not, the whole family is rigid. This is the exact sense of "top layer, carried by no section." [COMPLETE]

Part II. The Local Valuation Theorem

8 Setup and the parity lemma

A single context has an isotropic (internally commuting) label group $G \subseteq \mathbb{Z}_d^{2m}$, $\langle u, v\rangle \equiv 0 \pmod d$ for all $u, v \in G$. The Weyl operators define a $U(1)$-valued 2-cocycle on G,

$$f(u,v) = \tau^{\,c(u,v)}, \qquad W(u)W(v) = f(u,v)\, W((u+v) \bmod d),$$

with c as in Lemma 2. We say f *trivializes* (the context is *classical*) if there is a cochain $\mu : G \to U(1)$ with $f(u,v) = \mu(u)\mu(v)\mu(u+v)^{-1}$; then $V(v) := \mu(v)^{-1}W(v)$ is an honest (nonprojective) representation of G. The *valuation* of the context is the least N with an admissible μ valued in μ_N.

Lemma 26 (Parity lemma)**.** *For all u, v, $c(u,v) \equiv \langle u,v\rangle + d\cdot\rho(u,v) \pmod 2$ for an integer ρ; hence for even d, $c(u,v) \equiv \langle u,v\rangle \pmod 2$, and on a commuting pair at even d, $c(u,v)$ is even. For odd d the correction $d\,\rho$ may be odd, so $c(u,v)$ can be odd on a commuting pair.*

Proof. [COMPLETE] One qudit suffices (both sides add over sites). With $u = (a,b), v = (a',b')$ and carries $e_a, e_b \in \{0,1\}$ defined by $a + a' = (a + a' \bmod d) + d\,e_a$, $b + b' = (b + b' \bmod d) + d\,e_b$, Lemma 2 gives $c(u,v) = -ab - a'b' + 2a'b + q((u+v) \bmod d)$. Expanding $q((u+v) \bmod d) = (a + a' - de_a)(b + b' - de_b)$,

$$c(u,v) = ab' + a'b - d\big(e_b(a+a') + e_a(b+b')\big) + d^2 e_a e_b.$$

Mod 2, $ab' + a'b \equiv ab' - a'b = \langle u,v\rangle$, so $c(u,v) \equiv \langle u,v\rangle + d\rho \pmod 2$ with $\rho = e_b(a+a') + e_a(b+b') + d\,e_a e_b$. For even d the $d\rho$ term vanishes mod 2; on a commuting pair $d \mid \langle u,v\rangle$ with d even forces $\langle u,v\rangle$ even, so $c(u,v)$ is even. $\square$

Corollary 27 (Even-d contexts need no half-τ steps)**.** *For even d, every closed commuting context has* even *τ-level phase: its product is an ω-power $\omega^{s(C)}I$, $s(C) \in \mathbb{Z}_d$, with integrality automatic (not a gauge choice). This is Lemma 13, now seen as an immediate consequence of Lemma 26: the telescoped exponent $T = \sum_i c(P_i, v_i)$ is a sum of even integers.* [COMPLETE]

9 The cyclic transgression

Lemma 28 (Cyclic power scalar)**.** *Let $g \in G$ have order $a = \mathrm{ord}(g)$ in $\mathbb{Z}_d^{2m}$ (so $a \mid d$). Then $W(g)^a = \tau^\Gamma I$ with*

$$\Gamma \equiv a(a-2)\,q(g) \pmod{2d}.$$

Consequently any trivializing μ must satisfy $\mu(g)^a = \tau^\Gamma$, and one may take $\mu(g) = \tau^{(a-2)q(g)}\zeta$ with $\zeta^a = 1$; the essential factor is $\tau^{(a-2)q(g)} \in \mu_{2d}$.

Proof. [COMPLETE] Telescoping (Lemma 3 applied to the word $g, g, \ldots, g$ of length a, whose sum is $ag \equiv 0$), $\Gamma = \sum_{k=1}^{a-1} c(kg \bmod d,\ g)$. Using $c(w,g) = -q(w) - q(g) + 2\beta(w,g) + q((w+g) \bmod d)$ with $w = kg \bmod d$: the terms $-q(kg) + q((k+1)g)$ telescope over $k = 1, \ldots, a-1$ to $q(ag \bmod d) - q(g) = -q(g)$; the constant $-q(g)$ sums to $-(a-1)q(g)$; and $2\sum_{k=1}^{a-1}\beta(kg \bmod d, g) \equiv 2q(g)\sum_{k=1}^{a-1} k = q(g)a(a-1) \pmod{2d}$ (as $\beta(g,g) = q(g)$ and reductions cost multiples of $2d$). Summing, $\Gamma \equiv -q(g) - (a-1)q(g) + a(a-1)q(g) = q(g)\big(-a + a(a-1)\big) = a(a-2)q(g) \pmod{2d}$. For a genuine rep $V(g) = \mu(g)^{-1}W(g)$ with $V(g)^a = V(ag) = I$ one needs $\mu(g)^a = W(g)^a = \tau^\Gamma$; the stated $\mu(g)$ solves this since $\mu(g)^a = \tau^{a(a-2)q(g)}\zeta^a = \tau^\Gamma$. $\square$

Lemma 29 (Divisibility / parity of the essential exponent)**.** *$(a-2)q(g)$ is* even *for every g when d is even; for odd d it can be odd.*

Proof. [COMPLETE] Since $a = \mathrm{ord}(g) \mid d$ and $a\,q(g) = \sum_i (a\,g_{a,i})g_{b,i} \equiv 0 \pmod{d}$ (each $a\,g_{a,i} \equiv 0$), we get $q(g) \equiv 0 \pmod{d/a}$. *Even d:* if a is even then $a-2$ is even and $(a-2)q(g)$ is even; if a is odd then d/a is even (as d is even, a odd), so $q(g)$ is even and again $(a-2)q(g)$ is even. *Odd d:* a is odd, $a-2$ is odd, and $q(g) \equiv 0 \pmod{d/a}$ with d/a odd allows $q(g)$ odd, so $(a-2)q(g)$ can be odd. □

10 The theorem

Theorem 30 (Local classicality, sharp valuation)**.** *Every single Weyl context (commuting label group G) is classical: its cocycle f is a coboundary. A trivializing cochain μ can be chosen valued in*

$$\mu_d \quad (d \text{ even}), \qquad \mu_{2d} \quad (d \text{ odd}),$$

and the odd-d bound is sharp: the group $\langle(1,1)\rangle$ at $d=3$ admits no μ_3-valued trivializer.

Proof. [COMPLETE] *Cyclic factors.* Decompose $G = \bigoplus_\ell \langle g_\ell \rangle$ (structure theorem for finite abelian groups). On each factor take $\mu(g_\ell) = \tau^{(a_\ell-2)q(g_\ell)}\zeta_\ell$ as in Lemma 28 and extend by $\mu(kg_\ell) = \mu(g_\ell)^k \tau^{-\Gamma_{\ell,k}}$, where $W(g_\ell)^k = \tau^{\Gamma_{\ell,k}} W(kg_\ell)$; this makes $V = \mu^{-1}W$ a representation of $\langle g_\ell \rangle$. By Lemma 29 the essential factor $\tau^{(a_\ell-2)q(g_\ell)}$ lies in μ_d for even d; and each $\Gamma_{\ell,k} = \sum_{j<k} c(jg_\ell, g_\ell)$ is a sum of *even* c-values (Lemma 26, commuting pairs, even d), so $\tau^{\Gamma_{\ell,k}} = \omega^{\Gamma_{\ell,k}/2} \in \mu_d$. Hence the whole cyclic cochain is μ_d-valued for even d, and μ_{2d}-valued for odd d.

Cross terms. The remaining data are the pairings f across distinct factors, cohomologous (via the global coboundary τ^{-q}, i.e. $W = \tau^{-q}V_0$ with $V_0 = X^a Z^b$ having cocycle ω^β) to the symmetric bilinear form $\beta|_G$ valued in $\mathbb{Z}_d$. A symmetric bilinear B on G has cocycle ω^B trivialized by any quadratic refinement $\psi : G \to \mathbb{Z}_d$ (resp. $\mathbb{Z}_{2d}$) with $B(u,v) = \psi(u) + \psi(v) - \psi(u+v)$; on $\bigoplus \langle g_\ell \rangle$ set $\psi(\sum_\ell k_\ell g_\ell) = \sum_\ell \psi_\ell(k_\ell) + \sum_{\ell<\ell'} k_\ell k_{\ell'} \beta(g_\ell, g_{\ell'})$, whose diagonal parts ψ_ℓ are the cyclic solutions above and whose cross parts are $\mathbb{Z}_d$-valued (so μ_d-valued in either parity). Symmetric-cocycle classes are additive over the cyclic decomposition ($\mathrm{Ext}(\bigoplus \mathbb{Z}_{a_\ell}, \mathbb{Z}_d) = \bigoplus \mathbb{Z}_{\gcd(a_\ell,d)}$), so the assembled μ trivializes f globally with the stated valuation.

Sharpness at $d=3$. Let $g = (1,1)$ on one qudit, $d=3$; then $a = \mathrm{ord}(g) = 3$, $q(g) = 1$, and Lemma 28 gives $\Gamma \equiv 3 \cdot 1 \cdot 1 = 3 \pmod{6}$, so $W(g)^3 = \tau^3 I = e^{i\pi} I = -I$ (directly: $(XZ)^3 = \omega^3 I = I$ at $d=3$ and $W(g)^3 = \tau^{-3}(XZ)^3 = -I$). Any trivializer obeys $\mu(g)^3 = W(g)^3 = -1$. But every element of μ_3 cubes to $1 \neq -1$; hence no μ_3-valued μ exists, while $\mu(g) = \tau \in \mu_6 = \mu_{2d}$ works. The valuation is exactly μ_{2d}. □

Status of the step. The cyclic factor, the parity lemma, the transgression/divisibility, and the $d=3$ sharpness are each [COMPLETE]. The cross-term assembly invokes the standard additivity of symmetric extension classes over a cyclic decomposition (finite abelian group cohomology), used as a named structural fact but elementary; we mark the assembly [COMPLETE] with that reliance made explicit.

Corollary 31 (Genuine τ-steps are an odd-d phenomenon)**.** *The half-τ ($\mu_{2d} \setminus \mu_d$) valuation occurs* only *for odd d, and there iff some cyclic factor has $(a-2)q(g)$ odd (equivalently $\mu(g)^a = -1$). For even d no context ever requires a half-τ step. This inverts the naive expectation that "more phase" lives at even d.* [COMPLETE]

Remark 32 (Repeated generators / nonminimal presentations). The valuation is an invariant of the pair (G, f), not of any generating set: μ is a function on the group G, and a redundant or nonminimal presentation only adds relations already satisfied by μ. Concretely, if a generator is repeated (or a relation $\sum n_\ell g_\ell = 0$ is imposed), the transgression consistency $\mu(0) = 1$ is automatic because Γ was computed from the intrinsic order $a = \mathrm{ord}(g)$; adding g twice to a context word multiplies the phase by $f(w, g)f(w + g, g)$, an even τ-power at even d (Lemma 26), so no new half-step appears. Hence the μ_d/μ_{2d} dichotomy is stable under all presentations. [COMPLETE]

11 Status

Each row records whether this document contains a self-contained proof ([COMPLETE]), a sketch resting on a named external result ([SKETCH: CITED]), or an explicit open sub-lemma ([GAP]). "Abstract item" points to the corresponding Paper B abstract claim.

Result (this document)	Status	Remark
Weyl product law (Lem. 2), telescope (Lem. 3)	[COMPLETE]	the sole external input, derived here
Thm. 4 Obstruction Spectrum ($2S \equiv 0$; $\{0, d/2\}$ / $\{0\}$)	[COMPLETE]	multiplicities, repeats, nonmaximal, disconnected all allowed (shown in proof)
Thm. 8 Commutator-carry form $b = \sum\langle u, v\rangle/d$	[COMPLETE]	*derived* from the cocycle, not asserted
Cor. 9 closed-form cycle criterion (Q)	[COMPLETE]	even-d; inter-context terms vanish by closedness
Prop. 10 exact $\mathbb{Z}_d$ vs. $\mathbb{Z}_2$ shadow	[COMPLETE]/[GAP]	soundness [COMPLETE]; Weyl-specific faithfulness of the shadow [GAP] (empirical)
Thm. 14 Detection equivalence (a)⇔(b)⇔(c)	[COMPLETE]	incl. composite-d SNF double-annihilator (Lem. 11)
Thm. 18 Value-bit formula	[SKETCH: CITED]	structure [COMPLETE]; exact carry coefficients via Lem. 17 + `Wformula.py`
Thm. 19(A) attainment, $d \equiv 2 \pmod 4$	[COMPLETE]	signed scaled Peres–Mermin, genuine $\mathbb{Z}_d$-certificate
Thm. 19(B) attainment, general even d	[SKETCH: CITED]	generality inherited from [1]
Thm. 19(C) explicit $d = 8, 16$ certificates	[SKETCH: CITED]	original; numerically matrix-verified
Thm. 24 Depth Rigidity (a),(b)	[COMPLETE]	top-layer meaning made precise; no lift carries the bit
Thm. 30 Local valuation μ_d/μ_{2d}, sharp	[COMPLETE]	incl. analytic $d = 3$ sharpness ($\mu(g)^3 = -1$)
Cor. 27, Cor. 31, Rem. 32	[COMPLETE]	no half-τ at even d; presentation-invariance

The most important remaining gap. The general even-d attainment for $d \equiv 0 \pmod 4$ (Theorem 19(B)) is the only *existence* statement not proved here by an elementary parametric construction; it rests on [1]. The elementary construction (A) covers $d \equiv 2 \pmod 4$ completely, and (C) gives explicit witnesses at $d = 8, 16$, but a uniform self-contained family for all $d \equiv 0 \pmod 4$ is not established in this document. Secondarily, Theorem 18's exact carry-coefficient bookkeeping

(Lemma 17) is computationally rather than hand-certified, and Proposition 10's claim that the $\mathbb{Z}_2$ carry shadow is *faithful* on Weyl families (not merely sound) is empirical. Everything else — the Spectrum Theorem, the carry form, detection equivalence with the composite-d argument, depth rigidity, and the sharp local valuation — is complete and self-contained.

References

[1] R. Raussendorf, C. Okay, M. Zurel, P. Feldmann, *The role of cohomology in quantum computation with magic states*, Quantum **7**, 979 (2023), arXiv:2110.11631.

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[4] P. Lisoněk, R. Raussendorf, V. Singh, arXiv:1401.3035 ($\mathbb{Z}_2$ parity-proof enumeration).

[5] M. Flores Gordillo, *The Obstruction Spectrum of Weyl Context Families and the 2-adic Tower*, working draft, 2026 (source of the certificate framework, `Wformula.py`, `verify_cert8.py`, `verify_cert16.py`, `mu_d_valuation.py`).